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WASHINGTON, D.C.
May 26-29, 2026
Pre-Conference: May 25-26, 2026

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The Benchmarking Process

From Initiation to Implementation

Start with Purpose

Building a Purpose: Key Concepts

Effective benchmarking begins with clear questions—not just answers. Because institutional data can be large and complex, simple, focused visuals are often more powerful for guiding inquiry than overly complex analyses.

A strong purpose should center on understanding your position within a peer group. This requires identifying the performance metrics most valued by leadership and institutional stakeholders.

Start with Purpose

Benchmarking should:

- **Compare performance across peer institutions**
- **Highlight strengths, gaps, and trends**
- **Incorporate meaningful aggregates**, such as peer averages, top performers, and distance-to-benchmark measures

Finally, metrics must align with institutional strategy. Selecting the right indicators ensures that benchmarking efforts directly support progress toward strategic goals.

Start with Purpose

Defining Benchmarking Metrics

Metrics should be driven by the report's purpose and the needs of its users. Department leaders and institutional stakeholders often provide valuable input on which IPEDS variables are most relevant.

Once a draft set of metrics is identified, IPEDS data dictionaries can be used to map those metrics to the appropriate variables. Integrated Postsecondary Education Data System provides comprehensive datasets and documentation to support this process.

Start with Purpose

Our Approach

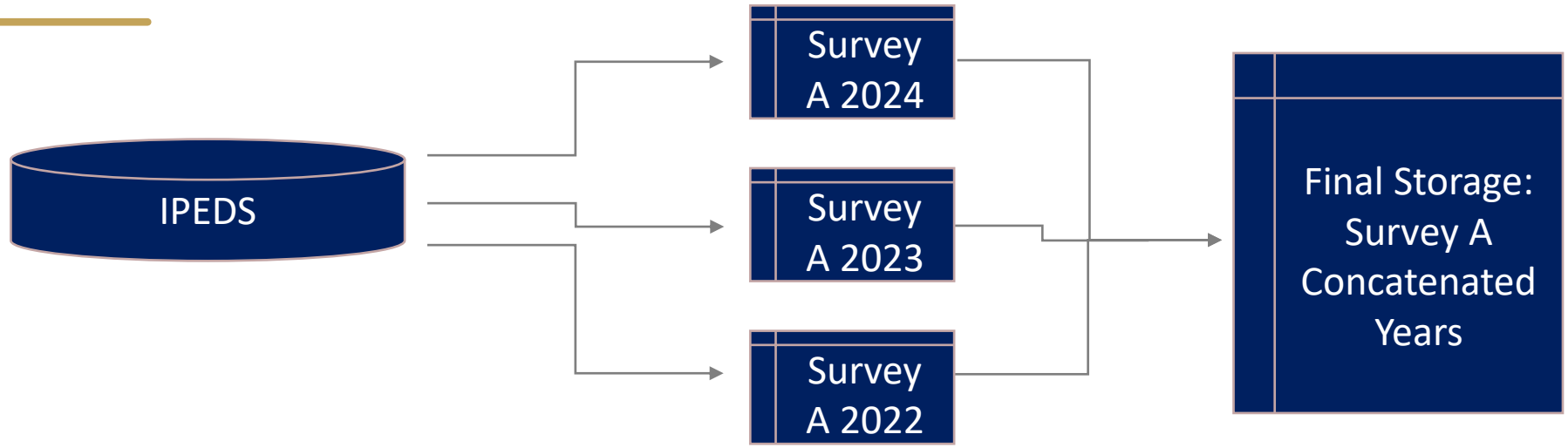
Our benchmarking framework was designed to support multiple institutional priorities:

- **Meeting accreditor requirements**
  Ensuring compliance with required standards and validating institutional performance using accreditor defined metrics.
- **Creating a centralized tool for tracking progress**
  Leveraging standardized, publicly available data to enable consistent use across departments and safe external sharing.

Start with Purpose

- **Staying aligned with national best practices**
  Identifying high-performing peer institutions to understand successful strategies.
- **Building peer-to-peer collaboration**
  Using benchmarking insights to foster communication and share actionable practices with peer institutions.
- **Supporting continuous improvement**
  Establishing an ongoing process to monitor performance and refine institutional strategy over time.

Create a Benchmarking Data Store



Each desired survey from IPEDS can be combined into one table for all collection years

Create a Benchmarking Data Store

Pull in Data

We used python to make requests for each dataset. Code for this can be found in the Github repository.

1. Define the year and dataset you want
2. Make the request to pull in the dataset
3. Store it in your preferred location
4. The end result for this step will be a table for each survey and year

Ex. EF2024A, EF2024B, EF2024C etc.

Create a Benchmarking Data Store

Structure the Data

This step will add the trending element to your report. For each survey table, you will:

1. Union the same survey across the different years that you pulled in.
2. Some surveys add new variables/columns from year to year, so you will have to add these variables to previous years as blank values.
3. We used python again to handle this structuring, but it can be as simple as starting with the most recent year and copy pasting the previous year to the dataset.

Ex. EF2024A union EF2023A union EF2022A union EF2021A etc.s

Create a Benchmarking Data Store

Structure the Data

UNITID	COLLECTION	EFALEVEL	LINE	SECTION	EFTOTLM
156107	2024	1	29	3	3136
156107	2023	1	29	3	2923
156107	2022_rv	1	29	3	2745
156107	2021_rv	1	29	3	2573

Create a Benchmarking Data Store

Structure the Data

Lastly, we put together our defined metrics using all needed surveys and collections.

1. If needed, different surveys can now be joined on UNITID (institution id) and collection year (self-defined).
2. Calculate the metric with the necessary IPEDS variables.
3. Some surveys may need to be prefiltered to certain codes. We made sure each join was only between tables that had one row per school per collection year.

Create a Benchmarking Data Store

Example Calculations

Source: EF20##D

First-Year Retention Rate

$$\frac{\text{RET_NMF} + \text{RET_NMP}}{\text{RRFTCTA} + \text{RRPTCTA}}$$

Source: C20##_B joined F2#2#_F1A

Outcome Efficiency Index

$$\frac{\text{CSTOTLT} * \$100{,}000}{\text{F1C191}}$$

Source: GR20##

Graduation Rate

$$\frac{\text{GRTOTLT (GRTYPE = 3)}}{\text{GRTOTLT (GRTYPE = 2)}}$$

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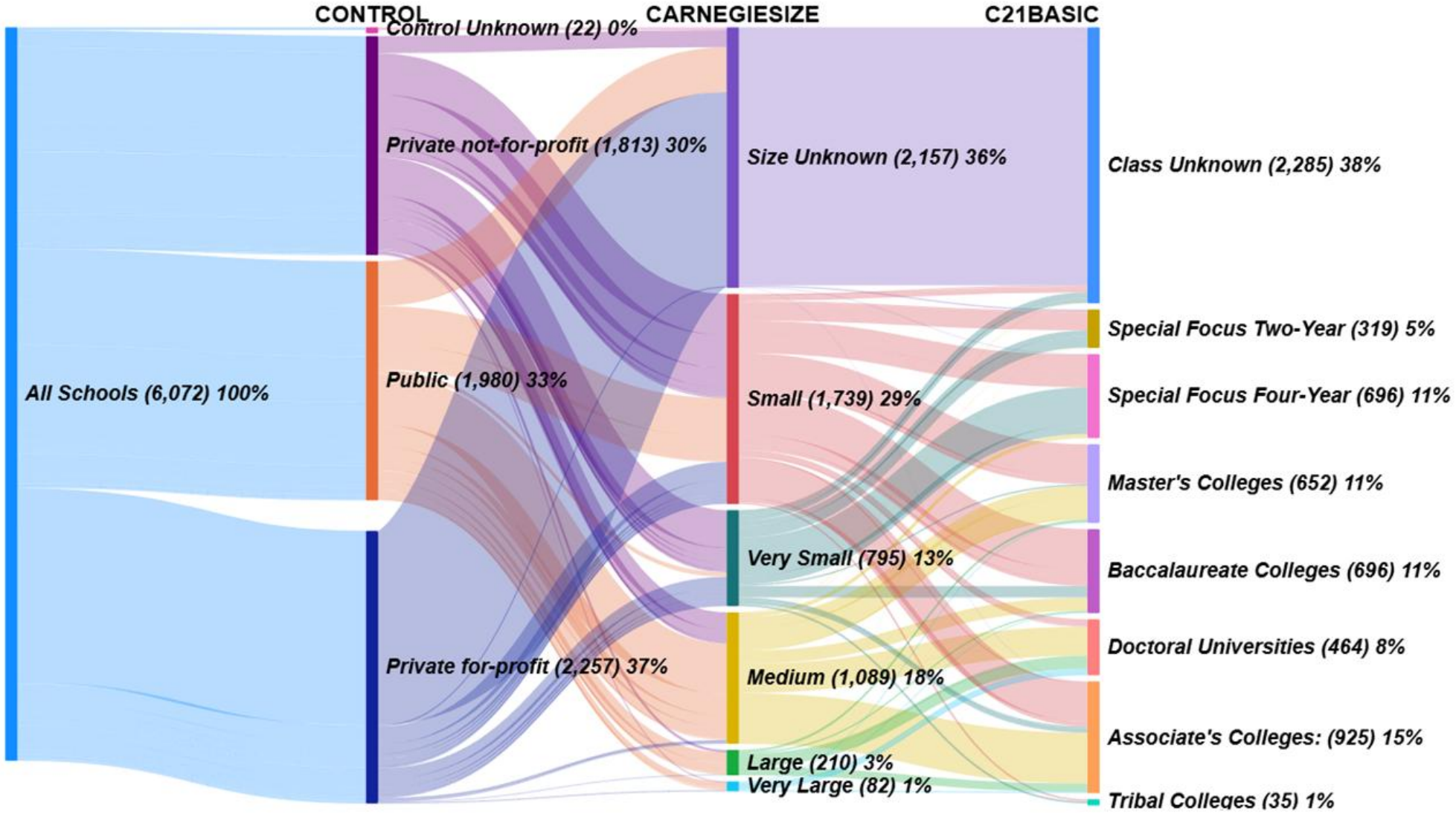
Create a Benchmarking Data Store

End Result

Institution ID	Year	First-Year Retention Rate	Outcome Efficiency Index	150% Graduation Rate
156107	2024	0.5641025641	2.96561973795	0.43
156107	2023	0.5576619273	3.28290891027	0.36
156107	2022	0.5082352941	3.28651778775	0.3
156107	2021	0.4727592267	3.22737016641	0.38

Grouping Peers



Grouping Peers

Filter down to similar institutions

Before we choose peers, it is helpful to establish a baseline pool of institution to choose your peers from.

Carnegie classifications help determine what schools have high level characteristics with your own school.

We used:

- Public vs Private (Control)
- Institution Size (CARNEGIESIZE)
- College Type (C21BASIC)

Institutional characteristics can be found in the directory survey from IPEDS (HD2024)

Grouping Peers

Example

We are:

- Public
- Medium to Very Large
- Associate's Colleges

Resulting pool of colleges had 444 to choose from.

Grouping Peers (cont)

Augment the selection process (optional)

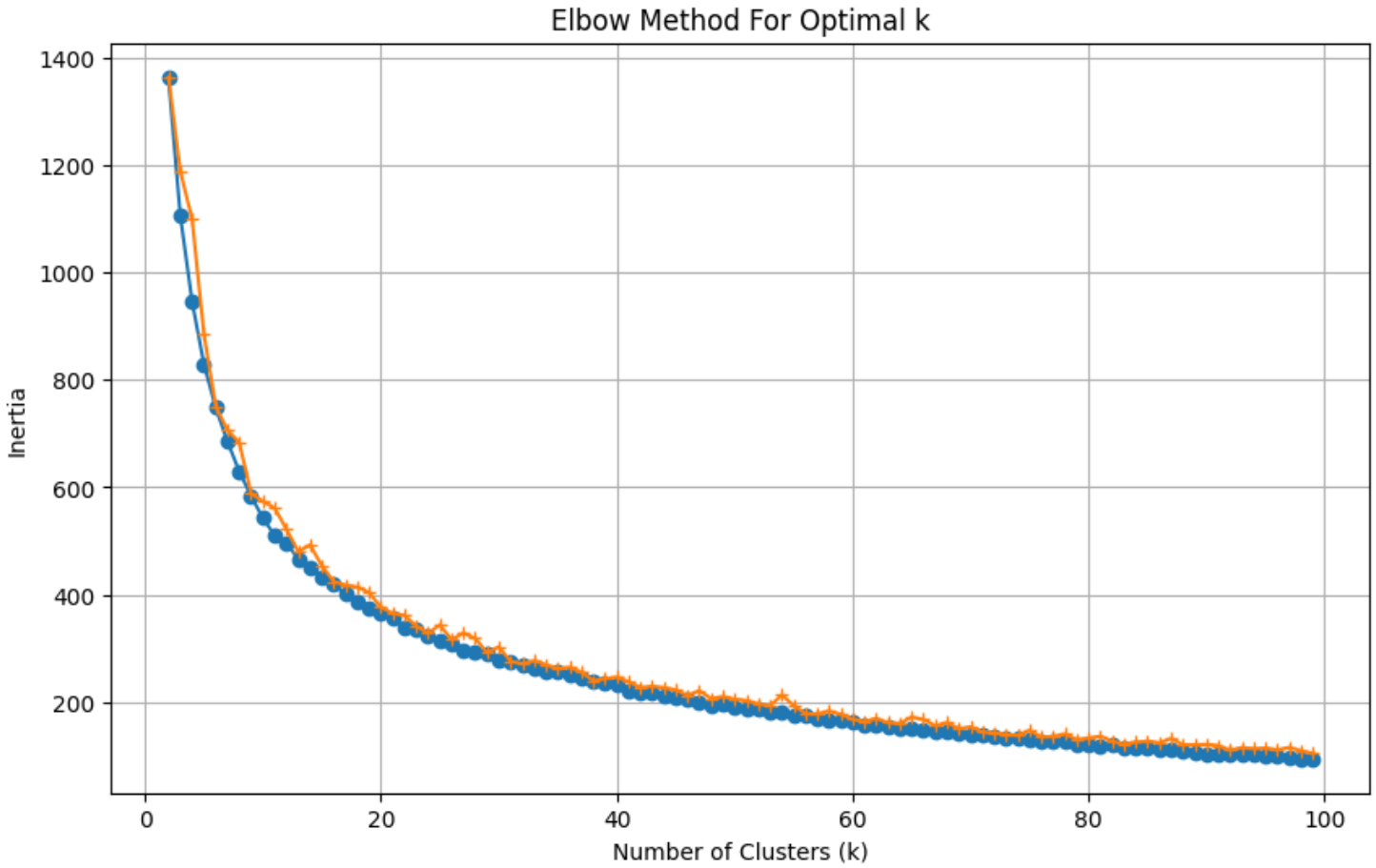
Using the baseline pool filtered down, we can also include clusters to help us understand potential peers.

K Means Clustering and a selection of metrics will create clusters to further group institutions with similar performance.

The code used for this will be in the Github Repository for this project. We used the Inertia and the Silhouette Method to find the optimal K number of clusters, which in our case was eight.

Grouping Peers (cont)

Augment the selection process (optional)

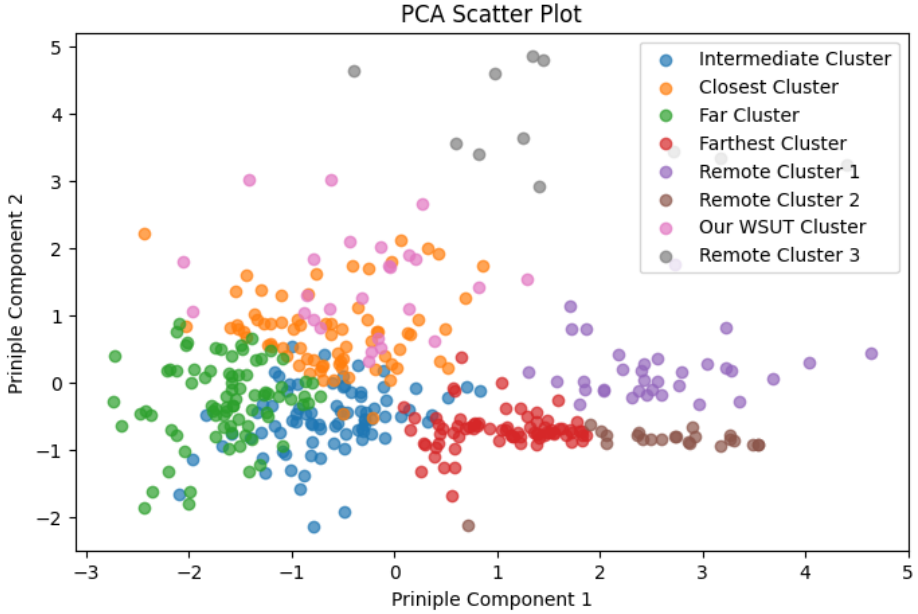


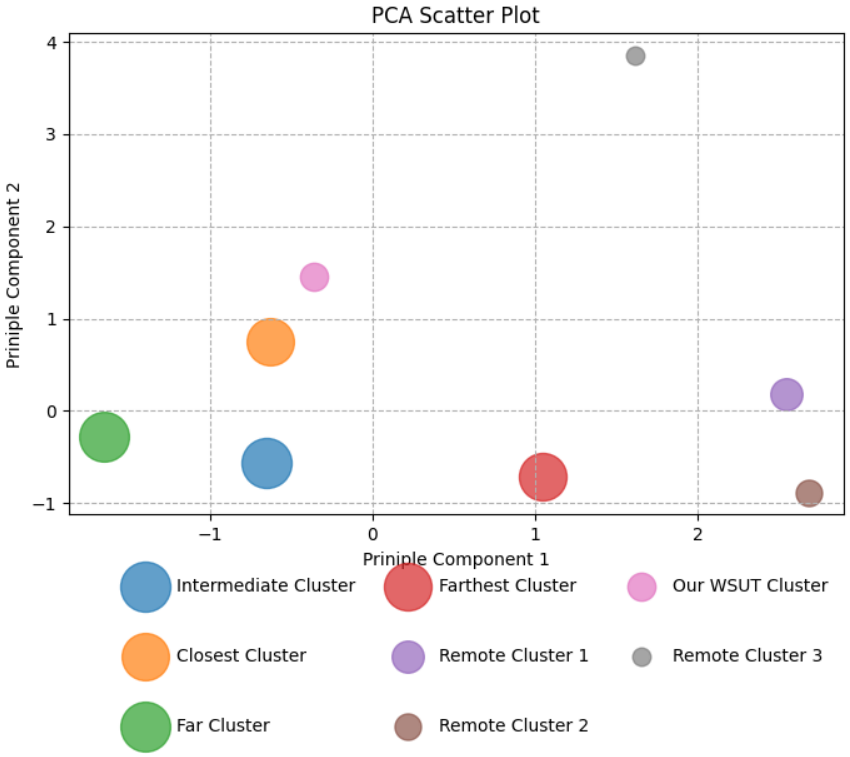
Grouping Peers (cont)

Augment the selection process (optional)

Next, we found the Principal Components of the same dataset (same selected metrics) to reduce the dimensions of our dataset. This is a preferred way to visualize the data to a 2-dimensional plane.

The first scatter plot are all schools graphed on the first two PCs, color coded to their corresponding cluster. The second is the average value for each cluster.





The Clusters were labeled after visually confirming their relationship with our own cluster.

Selecting Peers

Criteria

We used five metrics to base our selection criteria from.

1. First-Year Retention Rate
2. Graduation Rate
3. Completion & Transfer Rate
4. Outcome Efficiency Index
5. Fall Enrollment

Using theses five, we then defined the performance needed to be placed in our peer groups.

Selecting Peers

Criteria

Benchmarks

+/- 5% of WSU Tech
- Retention, Graduation, Completion & Transfer

+/- 1 of WSU Tech
- Outcome Efficiency Index

3,000+
- Fall Enrollment

Short-term Goals

+6%-12% of WSU Tech
- Retention, Graduation, Completion & Transfer

+1 of WSU Tech
- Outcome Efficiency Index

6,499+
- Fall Enrollment

Aspirational Goals

+13%-15% of WSU Tech
- Retention, Graduation, Completion & Transfer

+2 of WSU Tech
- Outcome Efficiency Index

6,500+ 3-year average
- Fall Enrollment

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Selecting Peers

Cosine Similarity (optional)

Cosine Similarity also helps us understand schools with similar or non-similar performance, but on a school-by-school basis. In this way, each school will have a distance score related to your own institution.

1. Using the same metrics again and baseline institution pool, create a new column that holds the cosine similarity (code used can be found in the github repo).
2. Order the new column in descending order. Cosine similarity is between 1 and -1, from most similar, to having opposite similarity, respectively. This can also be ordered by the different clusters created.

This extra information comes in handy when selecting multiple peers that meet your criteria. Using similarity helps narrow down your selection.

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Developing a Report

What you should have so far

- A **clear purpose** to guide report structure and content.
- **Defined metrics** sourced from IPEDS definitions. This may also include metric groupings if created.
- A validated **peer list** for comparison. This also could contain the **peer groups** that align with your institutional strategy.
- Supporting **institutional reference data** (e.g., labels, categories). This is found in the directory survey in IPEDS (HD2024)

Developing a Report

Displaying multiple dimensions

This is a common problem for data professionals to handle. How do you display as many factors or dimensions of data as possible in one visual? Here we have:

1. Time (Year over Year collections)
2. Peers (Multiple institutions)
3. The Metric itself
4. Benchmarking/Ranking/Comparison element

And not to mention any other breakdown you may need. One of many solutions we implemented was to have two dedicated pages that broke these dimensions apart:

- **Single Year Comparison**
- **Yearly Trend Comparison**

Developing a Report

Single Year Comparison

Here we visualize:

- Metrics in columns
- Peers in rows
- Added aggregate rows are:
    - Ranking for each metric that our school falls into
    - Change in rank from prior year

Developing a Report

Single Year Comparison

Select Year: 2024

Institution Name	Name and State		Student Success: First-Year Retention Rate	Student Success: Completion & Transfer Rate	Student Success: Graduation Rate	Financial Indicators: Outcome Efficiency Index	Financial Indicators: Composite Financial Index	Financial Indicators: Expenditure per FTE Student	Student Landscape: Student-to-Faculty Ratio	Student Landscape: Weighted Market Share Index	Student Landscape: Fall Enrollment
Benchmarks — Rank (1 2 3 4 5 6 7)			4	4	3	5	1	4	2	5	3
Change from Prior Year			↑ 1	↓ -1	↑ 1	↓ -2	→ 0	→ 0	→ 0	→ 0	↑ 2
WSU Tech	Wichita	KS	56%	68%	43%	3.0	9.1	$14,197	16	0.53	6,106
Western Iowa Tech Community College	Sioux City	IA	61%	71%	50%	2.0	0.4	$23,312	14	0.44	5,482
Tri-County Technical College	Pendleton	SC	54%	74%	42%	2.6	5.9	$14,860	20	2.35	5,890
Northshore Technical Community College	Lacombe	LA	53%	68%	40%	4.3	-2.7	$12,207	42	0.56	5,975
Lee College	Baytown	TX	58%	65%	53%	4.4	4.7	$16,988	31	0.96	8,323
John C Calhoun State Community College	Tanner	AL	59%	56%	41%	3.1	4.0	$13,318	22	0.97	9,119
Edison State Community College	Piqua	OH		64%	38%	3.4	-0.7	$10,208	23	0.09	4,624
Short Term Goals — Rank (1 2 3 4 5 6 7 8 9)			9	3	8	3	1	4	4	6	7
Change from Prior Year			→ 0	↑ 2	→ 0	→ 0	→ 0	↑ 1	→ 0	↑ 1	→ 0
WSU Tech	Wichita	KS	56%	68%	43%	3.0	9.1	$14,197	16	0.53	6,106
Lake Land College	Mattoon	IL	68%	66%	58%	2.9	7.8	$19,496	13	1.47	3,960
George C Wallace State Community Coll...	Hanceville	AL	61%	66%	51%	2.8	0.8	$15,824	18	0.66	6,424
Central Community College	Grand Island	NE	63%	67%	48%	1.6	4.2	$27,661	16	0.52	6,927
North Central Texas College	Gainesville	TX	60%	69%	35%	1.8	-0.5	$13,629	19	0.96	8,106
Gateway Community and Technical Colle...	Florence	KY	68%	59%	59%	5.7		$11,552	20	0.84	5,320
Elizabethtown Community and Technical ...	Elizabethtown	KY	65%	56%	59%	5.4		$10,633	26	0.48	6,420
Chippewa Valley Technical College	Eau Claire	WI	66%	63%	50%	2.5	-0.5	$22,663	14	1.01	8,330
Fox Valley Technical College	Appleton	WI	70%	77%	65%	1.9	2.6	$31,149	12	0.11	10,670
Aspirational Goals — Rank (1 2 3 4 5 6 7 8)			5	3	5	2	1	2	1	5	7
Change from Prior Year			↑ 1	↑ 1	→ 0	→ 0	→ 0	→ 0	→ 0	→ 0	→ 0
WSU Tech	Wichita	KS	56%	68%	43%	3.0	9.1	$14,197	16	0.53	6,106
Northcentral Technical College	Wausau	WI	53%	65%	45%	2.1	4.1	$26,490	18	0.07	6,746
Salt Lake Community College	Salt Lake City	UT	50%	54%	29%	1.5	6.3	$18,194	16	1.13	27,437
William Rainey Harper College	Palatine	IL	71%	68%	48%	1.7	3.2	$21,423	21	0.71	13,988
Owensboro Community and Technical C...	Owensboro	KY	64%	59%	54%	5.5		$12,084	21	0.42	4,875
Northeast Wisconsin Technical College	Green Bay	WI	63%	66%	51%	1.9	0.6	$27,948	17	0.39	11,826
Grand Rapids Community College	Grand Rapids	MI	54%	50%	21%	1.5	0.8	$17,834	19	0.54	12,468
Georgia State University-Perimeter College	Atlanta	GA	63%	85%	22%	1.3		$14,697	23	0.69	17,437

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Developing a Report

Yearly Trend Comparison

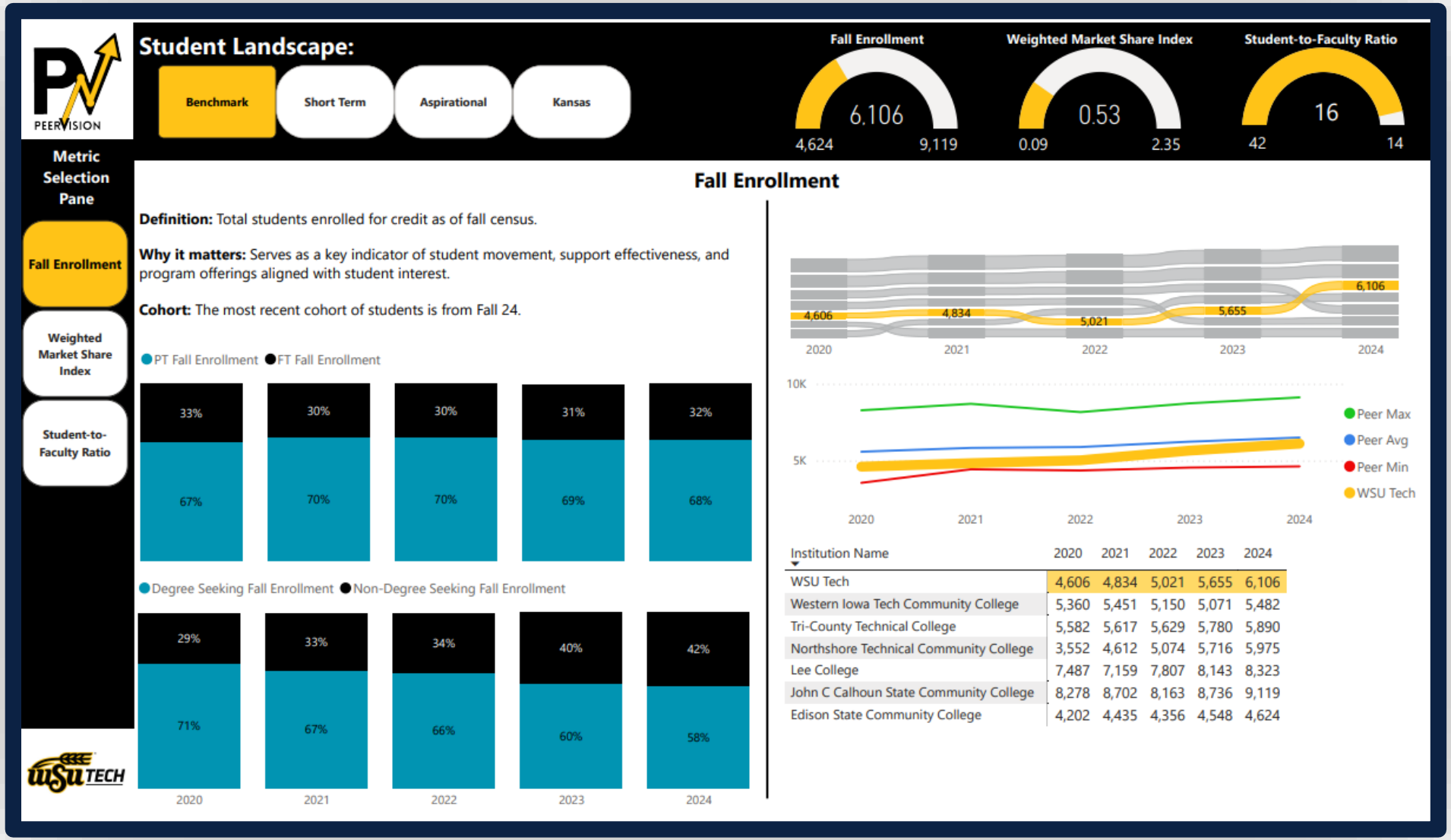
Here we visualize:

- Metrics typically on the y axis
- Peers as color codes or represented as an aggregate group
- Time of course lays on the x axis for each visual

You will also notice not all visuals are related to trend. These are not meant to be hard rules, but as helpful thought exercises to get a report that displays your benchmarking data in a meaningful way.

Developing a Report

Yearly Trend Comparison



Institution Name	2020	2021	2022	2023	2024
WSU Tech	4,606	4,834	5,021	5,655	6,106
Western Iowa Tech Community College	5,360	5,451	5,150	5,071	5,482
Tri-County Technical College	5,582	5,617	5,629	5,780	5,890
Northshore Technical Community College	3,552	4,612	5,074	5,716	5,975
Lee College	7,487	7,159	7,807	8,143	8,323
John C Calhoun State Community College	8,278	8,702	8,163	8,736	9,119
Edison State Community College	4,202	4,435	4,356	4,548	4,624

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Appendices

WSU Tech Metric Definitions

Notes on IPEDS Variables and Codes

Metric Definitions

First-Year Retention Rate

$$\frac{\text{RET_NMF} + \text{RET_NMP}}{\text{RRFTCTA} + \text{RRPTCTA}}$$

IPEDS Surveys: **EF20##D**

Metric Definitions

Completion & Transfer Rate

$$\frac{\text{OMCERT8} + \text{OMASSC8} + \text{OMBACH8} + \text{OMENRAI}}{\text{OMACHRT}}$$

IPEDS Surveys: **OM20##**

Notes: Filter **OMCHRT = 50**

Metric Definitions

Graduation Rate

$$\frac{\text{GRTOTLT (GRTYPE = 3, 30)}}{\text{GRTOTLT (GRTYPE = 2, 29)}}$$

IPEDS Surveys: **GR20##**

Notes: **GRTYPE** is 3 or 30 for 2-year or 4-year schools respectively in the numerator.

And then 2 or 29 for the denominator. In this case it was easiest for us to make two separate joins with the same table, one for the top and another for the bottom. Then make the division calculation. For more information, consult the IPEDS dictionary for GR2024.

Metric Definitions

Outcome Efficiency Index

$$\frac{\text{CSTOTLT} * \$100{,}000}{\text{F1C191}}$$

IPEDS Surveys: **C20##_B + F2#2#_F1A**

Notes: The number of distinct students who have completed per $100,000 of total expenses in the same FY.

Metric Definitions

Composite Financial Index

IPEDS Surveys: **F2#2#_F1A**

Calculation:

1. Primary Reserve Ratio (35%)

$$PRR = \frac{F1A17 + F1A15}{F1D02}$$

$$C_1 = 0.35 \times \left(\frac{PRR}{0.133}, -4{,}10\right)$$

Metric Definitions

Composite Financial Index

2. Operating Revenue Ratio (10%) ORR

Loss Expenses

≤ 2019: $F1B27 - F1C191$

> 2019: $F1N01$

Revenue

≤ 2019: $F1B27 - F1C19IN$

> 2019: $F1N02$

$$ORR = \frac{\text{Loss Expenses}}{\text{Revenue}}$$

$$C_2 = 0.10 \times \text{clip}\left(\frac{ORR}{0.013}, -4{,}10\right)$$

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Metric Definitions

Composite Financial Index

3. Return on Net Assets (20%)

$$RONA = \frac{F1D03}{F1D04}$$

$$C_3 = 0.20 \times \text{clip}\left(\frac{RONA}{0.02}, -4{,}10\right)$$

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Metric Definitions

Composite Financial Index

4. Viability Ratio (35%)

$$VR = \frac{F1N05}{F1A10}$$

$$C_4 = 0.35 \times \text{clip}\left(\frac{VR}{0.417}, -4{,}10\right)$$

If viability inputs are missing, **maximum score is assigned**.

Composite Financial Index = C1+C2+C3+C4

Notes: IPEDS codes and variables can change collection to collection. **ORR** was a case where F1N01 was a replacement variable in 2020, so you need to make some adjustments to keep consistent measurements.

Metric Definitions

Expenditure per FTE Student

$$\frac{\text{F1C191}}{\text{EFTEUG}}$$

IPEDS Surveys: : **F2#2#_F1A + EFIA20##**

Notes: **EFTEUG** is the estimated full-time equivalent (FTE) undergraduate enrollment.

Metric Definitions

Student-to-Faculty Ratio

STUFACR

IPEDS Surveys: **EF20##D**

Notes: IPEDS uses their own definition for FTE students and faculty so make sure to consult the EF2024D or relevant collection definition.

Metric Definitions

Weighted Market Share Index

$$\frac{\text{EFRES02}}{\text{HS Grads} * \left(\frac{\text{DS}}{\text{SUM(DS)}}\right)}$$

IPEDS Surveys: **EF20##C + DRVEF20##**

Common Core Data: *https://nces.ed.gov/ccd/elsi/*

Notes: EFRES02 is the count of recently graduated high school students, and then we also filtered it to the institutions state code.

HS Grads comes from the Common Core Data site using the *Public School Student Enrollments by Grade* table. For each institution, we subset the high schools in a 25-mile radius

Metric Definitions

Weighted Market Share Index (cont)

Notes: **HS Grads** comes from the Common Core Data site using the *Public School Student Enrollments by Grade* table. For each institution, we subset the high schools in a 25 mile radius from the institution, and then added the HS membership for each AY to get the 12th grade counts. We then multiplied this by the state average HS graduation rate to proxy the number of available graduates in for the following Fall after their 12th grade year.

Metric Definitions

Weighted Market Share Index (cont)

Notes: Lastly, **DS** is **EFUG1ST + EFUGTRN + EFUGCNT** from the DRVEF20## survey in IPEDS. It is the total number of degree seeking students, which we filtered to only the institutions within a 25-mile radius of the target institution. The numerator is the target institutions DS and the denominator is the total local DS enrollment. The product of this ratio and **HS Grads** ensures that a school is indexed weighted on their local enrollment size. If there is only one college in the area, they will have a higher recent HS graduate enrollment than a college that has multiple local competitors.

Metric Definitions

Fall Enrollment

ENRTOT

IPEDS Surveys: **DRVEF20##**

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Notes on IPEDS Variables and Codes

Inconsistencies

Some surveys like the finance **F2020_F1A** will update and add/delete variables. Another change could be the codes being used year to year. In a few of our metrics, we use the codes to filter to total students or some subset, but these codes are liable to change to mean something different. However, this may not be immediately evident, especially for unfamiliar institutions. So it is beneficial to go through each collection definition being used for your own created metric to make sure you are staying consistent throughout your collection years.